# Molecular Systems Biology and Dynamics: An Introduction for non-Biologists

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**Abstract**

This document provides an introduction to basic molecular systems biology concepts, describing several questions in dynamics that arise in the field.

## 1 Introduction

Within the last few years, the field of "molecular systems biology" has taken shape, having as its goal the unraveling of the basic dynamic processes, feedback control loops, and signal processing mechanisms underlying life at the cellular level. Leading biologists have recognized that new systems-level knowledge is urgently required in order to conceptualize and organize the revolutionary developments taking place in the biological sciences, and new academic departments and educational programs are being established at major universities.

The studies of dynamics, feedback, and signal processing in biology have been seriously pursued for at least 50 years, in the fields of biological and biomedical engineering, biomathematics, and biophysics, This has resulted in quantitative theories of physiological regulation, metabolic pathways, insulin control, heart electrical patterns, neural and circadian oscillations, and so forth. In related work, ecology and population dynamics have dealt with many similar problems but at a larger scale.

So, one may ask, why the sudden resurgence of interest? The answer surely involves a combination of many factors. Bioinformatics has been tremendously successful in facilitating the sequencing of human, animal, plant, bacterial, and other genomes, as well as in protein structure prediction. Nontrivial ideas and algorithms from discrete mathematics, probability and statistics, theoretical computer science, and even partially observed stochastic systems (Hidden Markov Models), embedded in user-friendly software, are now indispensable tools of the working biologist and pharmaceutical researcher. Thus, many biologists have come to accept and value the use of mathematical tools. On the other hand, new data collection and measurement approaches, themselves based upon sophisticated engineering, make possible the simultaneous monitoring of the activity of thousands of genes and the concentrations of proteins and metabolites, thus allowing for the study of microscopic dynamic interactions among cellular components, and making a systems-level view of cells particularly natural. The huge amounts of data being generated by genomics and proteomics require new theoretical approaches to interpretation and organization. Medical advances also drive this new emphasis. Many in the pharmaceutical industry have come to the realization that only by understanding cells as a whole can one identify novel targets for new drugs, and understand their systemic effects; gene therapies will depend on a more global understanding of dynamic interactions among genes and their cellular environment. Finally, and at a somewhat more philosophical level, there also is the fact that current experimental methods permit making *falsifiable predictions*, bringing modern biology closer to physics and chemistry as a science. While classical theoretical biology dealt largely with ecology, or with whole organisms, biologists can now test hypotheses in a precisely targeted fashion. For example, if a mathematical model predicts that a certain mutation will make fruit flies grow a leg instead of antennae on their heads, the mutation can be carried out and the results observed.

Here, we provide an introduction to many of the basic concepts of molecular biology.[1] and we describe some of the central questions in dynamics that arise in the field.

## 2 Molecular Cell Biology

The fundamental unit of life is the cell (Figure 1). Organisms may consist of just one cell or they may be

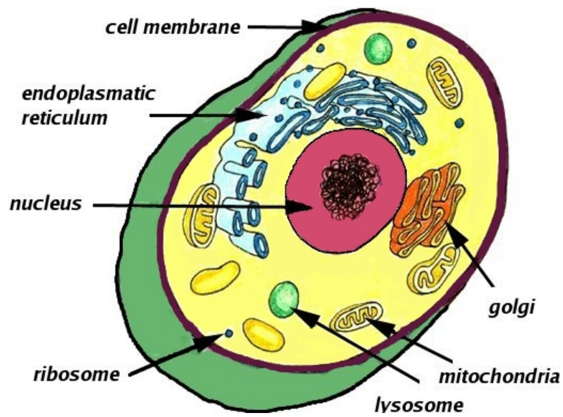


Figure 1: An eukaryotic cell

multicellular; the latter type are typically organized into tissues, which are groups of similar cells arranged so as to perform a specific function. (For example, humans have on the order of $10^{14}$ cells organized into roughly 200 tissues.)

One may view cell life as a collection of "wireless networks" of interactions among proteins, RNA, DNA, and small molecules involved in signaling and energy transfer. These networks process environmental signals, induce appropriate cellular responses, and sequence internal events such as gene expression, thus allowing cells and entire organisms to perform their basic functions. These control and communication networks can be relatively simple, such as the *two-component systems* found mainly in bacteria, which are cascades connecting sensors (proteins in the cell membrane, which detect outside signals) to actuators (typically transcription factors, which direct the expression of a gene), cf. [27]. Or they may be incredibly sophisticated, as in higher organisms, involving multiple *signal transduction pathways* in which information is relayed among enzymes through chemical reactions (for instance, phosphorylation).

In addition to their own needs for survival and reproduction, cells in multicellular organisms need additional levels of complexity in order to enable communication among cells and overall regulation, as well as to direct differentiation from a single fertilized egg into the various tissues in an individual member of a species. We will focus on intracellular pathways, but these other aspects are no less exciting areas of study.

Before providing more details and examples, let us step back and review some of the basic concepts and terminology.

[1]For a serious study of the subject, a recommended starting point is the book [1]

## 2.1 Prokaryotes, Eukaryotes, Archaea, and Viruses

At the highest level, biologists classify life forms into prokaryotes, eukaryotes, and archaea. *Prokaryotes* are organisms whose cells do not have a nucleus nor other well-defined compartments; their genetic information is stored in chromosomes –typically circular– as well as in smaller circular DNA molecules called plasmids. *Eukaryotes* have cells with organized compartments; their genetic material is stored in chromosomes –typically linear– that lie in the nucleus. Most prokaryotes, with few exceptions, are unicellular, and most are bacteria. Eukaryotes might be unicellular (e.g., yeast) or multicellular (e.g., plants and animals). *Archaea* were proposed as a third life form in the mid-1970s, and they share many characteristics with both prokaryotes and eukaryotes.

Eukaryotic cells are enclosed in a *plasma membrane*, which is made up of lipids and also contains proteins and carbohydrates, and acts as a protective barrier and gatekeeper, permitting only selected chemicals to enter and leave the cell. (In addition to membranes, plant cells also have a rigid cell wall.) Their interior is called the cytoplasm, and many types of organelles —specialized compartments— populate the cell (mitochondria, responsible for energy production through metabolism, and containing a very small amount of DNA; chloroplasts for photosynthesis; ribosomes, responsible for protein synthesis, and made up themselves of proteins and RNAs; endoplasmic reticulum; and so forth). The cytoskeleton, made up of microtubules and filaments, gives shape to the cell and plays a role in intracell substance transport. Prokaryotic cells, on the other hand, are surrounded by a membrane and cell wall, but do not contain the usual organelles.

*Viruses* consist of protein-coated DNA or RNA, and are not usually classified as living organisms, because they cannot reproduce by themselves, but rather require the machinery of a host cell in order to replicate. In particular, bacteriophages are viruses that infect bacteria.

## 2.2 Genomics and Proteomics

Research in molecular biology, genomics, and proteomics has produced, and will continue to produce, a wealth of data describing the elementary components of intracellular networks as well as detailed mappings of their pathways and environmental conditions required for activation.

### DNA and Genes

The *genome*, that is to say, the genetic information of an individual, is encoded in double-stranded *deoxyribonucleic acid (DNA)* molecules, which are arranged into chromosomes. It may be viewed as a "parts list" which describes all the proteins that are potentially present in every cell of a given organism. Genomics research has as its objective the complete decoding of this information, both the parts common for a species as a whole and the cataloging of differences among individual members.

The key paradigm of molecular biology: "DNA makes RNA, RNA makes protein, and proteins make the cell" is called the *central dogma of molecular biology* (Crick, 1958).[2] A separate process, *replication*, occurs more rarely, and only when a cell is ready to divide (S phase of mitosis, in eukaryotes), and results in the duplication of the DNA, one copy to be part of each of the two daughter cells. See Figure 2. The term *gene expression* refers to the process by which genetic information gets ultimately transformed into working proteins. The main steps are transcription from DNA to RNA, translation from RNA to linear amino acid sequences, and folding of these into functional proteins, but several intermediate editing steps usually take place as well.

[2] Recent work is forcing a rethinking of the roles of RNA and proteins. For example, prions appear to take advantage of a direct mechanism for protein replication: when a prion infects an organism, it interacts with wild-type –that is to say, normal– proteins, causing them to change their shape. For another example, until recently, RNA was not believed to be a direct player in cell control mechanisms, but now it is known that double-stranded RNA's (dsRNA's) can act, through the RNA interference (RNAi) effect, to disrupt ("turn-off" or "silence") genes. However, the central dogma remains the organizing principle: as usual in biology, the only general "theorem" is that every general fact has exceptions!

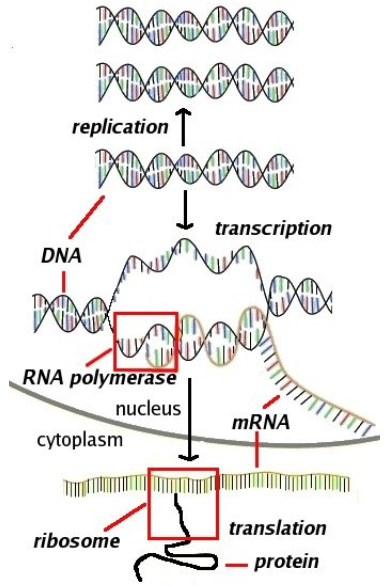


Figure 2: Central dogma of molecular biology

(Sometimes the term "gene expression" is used only for the transcription part of this process.) At any given time, and in any given cell of an organism, thousands of genes and their products (RNA, proteins) actively participate in an orchestrated manner.

The DNA molecule is a double-stranded helix made of a sugar-phosphate backbone and nucleotide bases (Figure 3). Each strand carries the same information, which is encoded in the 4-letter alphabet $\{A, T, C, G\}$

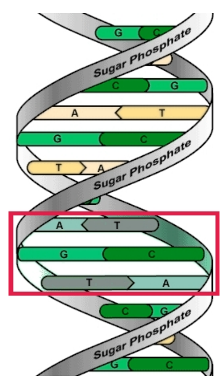


Figure 3: DNA; a codon shown in box

(the nucleotides Adenine, Thymine, Cytosine, and Guanine), in a "complementary" form ($A$ in one strand corresponds to $T$ in the other, and $C$ to $G$). The two strands are held together by hydrogen bonds between the bases, which gives stability but can be broken-up for replication or transcription. One describes the letters in DNA by a linear sequence such as:

`gcacgagtaaacatgcacttcccaggccacagcagcaagaaggaggaatc...`

and genes (instructions that code for proteins) are substrings of the complete DNA sequence. (Besides genes, there are regulatory and start/stop regions that help delimit genes as well as determine if and when they should be "active". In addition, there are also regions that have other roles, such as coding for RNA that may not lead to proteins.) Because of its double-stranded nature, DNA is chemically stable, and serves as a good depository of information. One might think of DNA storage as a "hard disk" in a vague computing analogy.

### RNA

The "read-out" of genetic information —bringing-in the instructions into working memory for execution, in our computer analogy— begins when DNA information is transcribed letter by letter into "RNA language." *Ribonucleic acid (RNA)* is a nucleic acid very similar to DNA, but less stable than DNA, and almost exclusively found in single-stranded form (with exceptions such as the RNA in some viruses). RNA language is basically the same as DNA's, with the minor (for us) detail that in RNA, the amino acid thymine is replaced with uracil, symbolized by the letter $U$. This process is known as *transcription*. The "copying-machine" is called *RNA polymerase*. A polymerase is, generally speaking, an *enzyme* —a type of protein that acts as a catalyst— that helps in the synthesis of nucleic acids. RNA polymerase is, thus, a polymerase that helps make RNA, more precisely *messenger RNA (mRNA)*.[3] A *promoter region* is a part of the DNA sequence of a chromosome that is recognized by RNA polymerase. In prokaryotes, the promoter region consists of two short sequences placed respectively 35 and 10 nucleotides before the start of the gene. Eukaryotes require a far more sophisticated transcriptional control mechanism, because different genes may be only active in particular cells or tissues at particular times in an organism's life; promoters act in concert with enhancers, silencers, and other regulatory elements.

### Proteins

*Proteins* are the primary components of living things. Among other roles, they form receptors that endow the cell with sensing capabilities, actuators that make muscles move (myosin, actin), detectors for the immune response, enzymes that catalyze chemical reactions, and switches that turn genes on or off. They also provide structural support, and help in the transport of smaller molecules, as well as in directing the breakdown and reassembly of other cellular elements such as lipids and sugars. Ultimately, one might say that cell life is about proteins and how and when they are produced.

After transcription, *translation* is the next step in the process of protein synthesis and it is performed at the *ribosomes*. The information in the mRNA is read, and proteins are assembled out of amino acids (with the help of *transfer RNA (tRNA)*, which help bring in the specific amino acids required for each position). RNA language is translated into protein language by a mapping from strings written in the RNA alphabet $\Sigma_{\rm n} = \{U, A, G, C\}$ into strings written in the amino acid alphabet:

$$\Sigma_{\rm a} = \{A, R, D, N, C, E, Q, G, H, I, L, K, M, F, P, S, T, W, Y, V\}.$$

Every sequence of three letters in the RNA alphabet $\Sigma_{\rm n}$ is replaced by a single letter in the alphabet $\Sigma_{\rm a}$. The genetic code explains how triplets (or *codons*, one of which is shown in Figure 3) of bases map into individual amino acids. The code, including full names and three and one-letter abbreviations, is shown in Figure 4. For example, the codon AUG translates into M (Methionine). Thus, the DNA string $TACTCATTGCGC$ would first get transcribed into the RNA string $AUGAGUAACGCG$ (note the complementation, and replacing $T$ by $U$), and would be then translated into the sequence $MSNA$ (Methionine-Serine-Asparagine-Alanine) of amino acids. The string AUG codes for the amino acid Methionine but also serves as a "start" codon: the first AUG in an mRNA indicates where translation should begin.

The *shape* of a protein is what largely determines its function, because proteins interact with each other, and with DNA and metabolites, through lego-like fitting of parts in lock and key fashion, transfer of small molecules, or enzymatic activation. Therefore, the elucidation of the three-dimensional *structure* of proteins

[3]This description is over-simplified: in eukaryotic cells, an intermediate form of RNA called heterogeneous nuclear RNA (hnRNA) is produced first; then a process of "editing" gets rid of "introns" which are not part of the code for the desired protein, leaving the "exons" that are joined together to produce the actual mRNA, perhaps after insertion of some additional nucleotides.

| | | | |
|---|---|---|---|
| Alanine Ala A | GCU, GCC, GCA, GCG | Leucine Leu L | UUA, UUG, CUU, CUC, CUA, CUG |
| Arginine Arg R | CGU, CGC, CGA, CGG, AGA, AGG | Lysine Lys K | AAA, AAG |
| Asparagine Asn N | AAU, AAC | Methionine Met M | AUG |
| Aspartic Acid Asp D | GAU, GAC | Phenylalanine Phe F | UUU, UUC |
| Cysteine Cys C | UGU, UGC | Proline Pro P | CCU, CCC, CCA, CCG |
| Glutamine Gln Q | CAA, CAG | Serine Ser S | UCU, UCC, UCA, UCG, AGU, AGC |
| Glutamic Acid Glu E | GAA, GAG | Threonine Thr T | ACU, ACC, ACA, ACG |
| Glycine Gly G | GGU, GGC, GGA, GGG | Tryptophan Trp W | UGG |
| Histidine His H | CAU, CAC | Tyrosine Tyr Y | UAU, UAC |
| Isoleucine Ile I | AUU, AUC, AUA | Valine Val V | GUU, GUC, GUA, GUG |
| START | AUG, GUG | STOP | UAG, UGA, UAA |

Figure 4: Genetic code

is a central goal in biochemical research; this subject is studied in the fields of *proteomics* and *structural biology*. The *Protein Data Bank* (http://www.rcsb.org/index.html) based at Rutgers University, USA, serves as an online catalog of protein structures. Sometimes, protein structure can be gleaned through physical methods, such as X-ray crystallography or NMR spectroscopy. Very often, however, the structure of a protein P can only be estimated, based upon a comparison with an *homologous* protein Q whose structure has been already determined (as chemists say, "solved"). One says that P and Q are homologous if they are, in an appropriate sense, close in amino acid sequence, or equivalently, in the DNA sequences for the genes coding for P and Q. One measure of closeness is Hamming distance (by how many "letters" do P and Q differ?), but more sophisticated measures used in practice include allowance for deletions and insertions of letters in P and Q. The rationale behind homology-based protein shape determination is that homologous proteins probably share a common evolutionary or developmental ancestry, and hence perform similar functions. Mathematical methods of computational biology (bioinformatics) play a central role in homology approaches; the *critical assessment of structure prediction methods (CASP)* competition compares methods from different researchers. Yet another set of techniques for elucidating the shape of proteins from their description as a linear sequence of amino acids is that of *energy minimization methods*. One views the protein-folding process as a gradient dynamical system, of which steady states are the stable configurations. This method is very difficult to apply, because of the complexity of the energy function, but has been useful for comparatively small proteins.

After translation, proteins are typically subjected to *post-translational modifications*, such as the addition of phosphate or methyl groups, or, in eukaryotic cells, *ubiquitination*, the process by which a protein is inactivated by attaching ubiquitin to it. Ubiquitin is a protein whose function is to mark other proteins for *proteolysis* (degradation), a process which occurs at the *proteasome*.

One of the key properties of proteins is that their shape (conformation) can be modified in a predictable fashion, as the consequence of interactions with other molecules. One often says that the protein has been "activated" as a result of such an interaction. For instance, Figure 5 shows, in schematic form, two conformations of the recoverin protein, the second of which comes about when two calcium ions have been inserted at appropriate places (white balls). Notice how the insertion of these ions makes an "arm" swing out. Depending on the position (extended or not) of this arm, different interactions of this protein with other players in the cell will occur.

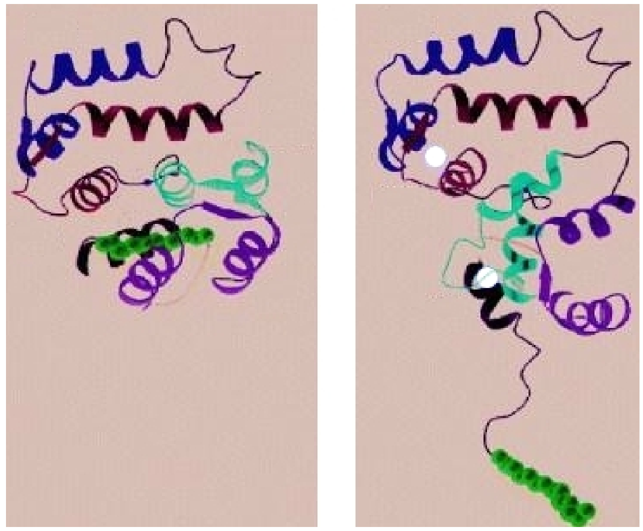

Figure 5: A protein in two conformations. Left one is $Ca^{2+}$-free. Right one is $Ca^{2+}$-bound

### 2.3 Proteins act as Sensors, Signal Relayers, and Actuators

Conformation changes in proteins typically happen in response to intracellular or extracellular ligand binding events, or because of binding with other proteins. (To *bind* means to reversibly join; *ligands* are small molecules that bind with larger molecules, typically proteins.) Two noteworthy instances of activation are provided by receptors and by phosphorylation reactions.

*Receptors* are proteins that act as the cell's sensors of outside conditions, relaying information to the inside of the cell. A receptor is typically made up of three parts. The *extracellular domain* ("domains" are parts of a protein) is exposed to the exterior of the cell. Extracellular ligands, such as growth factors and hormones, bind to receptors, most of which are designed to recognize a specific type of ligand. The *transmembrane domain* serves to "anchor" the receptor to the membrane. Finally, a *cytoplasmic domain* helps initiate reactions inside the cell in response to exterior signals, by interacting with other proteins. There is a special class of receptors which constitute a common target of pharmaceutical drugs: *G-protein-coupled receptors (GPCR's)* (Figure 6). The name of these receptors arises from the fact that, when their conformation changes in response to a ligand

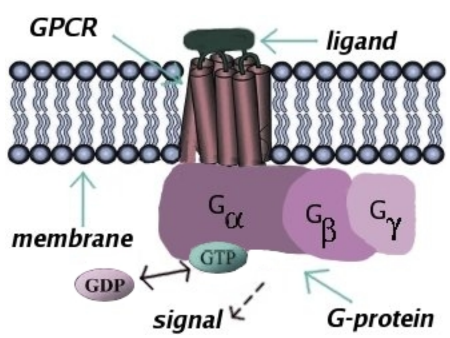


Figure 6: G-protein-coupled receptor and G-protein

binding event, they activate G-proteins, so called because they employ *guanine triphosphate* and *diphosphate (GTP* and *GDP)* in their activity. GPCR's are made up of several subunits ($G_\alpha$, $G_\beta$, $G_\gamma$) and are involved in the detection of metabolites, odorants, hormones, neurotransmitters, and even light (rhodopsin, a visual pigment).

Another example of activation is *phosphorylation*. *Adenosine triphosphate (ATP)* is a nucleotide that is the major energy currency of the cell. An *enzyme* is a protein that catalyzes a chemical reaction. Phosphorylation is a chemical reaction in which an enzyme X —called a *kinase* when playing this role— transfers a phosphate group ($PO_4$) from a "donor" molecule such as ATP to another protein Y, which becomes "activated" in the sense that its energy is increased. Once activated, protein Y may then influence other cellular components, including other proteins, itself acting as a kinase, or it may take an appropriate shape that allows it to to bind with yet another protein or to a segment of DNA so as to initiate, enhance, or repress expression of a gene. Normally, proteins do not stay activated forever; another type of enzyme, called a *phosphatase*, eventually takes away the phosphate group; see Figure 7. In this manner, signaling is "turned off" after a while, so that the system is

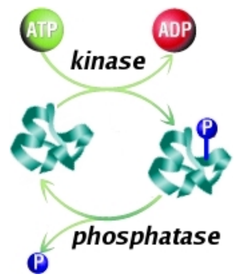


Figure 7: Phosphorylation and de-phosphorylation

ready to detect new signals.

Receptors and enzymatic cascades act in concert. Binding of extracellular ligands triggers signaling through a series of chemical reactions inside the cell, carried out by enzymes and often relayed by smaller molecules called *second messengers*. In this manner, regulatory pathways can be either turned "on" and "off" or modulated, and transcription of particular sets of genes may be started and stopped in response to environmental conditions. Figure 8 ([6]) illustrates one such pathway, which involves GPCR activation as well as signaling through a MAPK cascade (more on MAPK cascades below).

The animation at *http://biocreations.com/pages/mapk.html* is strongly recommended as an illustration of signaling pathways.

As another illustration, consider the diagram shown in Figure 9, extracted from the paper [12] on cancer research, describing the top-level schematics of a wiring diagram of signaling circuitry in the mammalian cell. The illustration shows the main signaling pathways for growth, differentiation, and apoptosis (commands which instruct the cell to die). Highlighted in red are some of the genes known to be functionally altered in cancer cells. Of course, such a figure, compared for example with the more detailed biochemical pathway shown in Figure 8, leaves out a lot of information, some known but omitted for simplicity, and some unknown. Much of the system has not been identified yet, and the functional forms of the interactions, much less parameters, are only very approximately known. However, data of this type are being collected at an amazing rate, and better and better models are being obtained constantly.

Both of the above examples were from eukaryotes. We now turn to one from a prokaryote. *Chemotaxis* is the term used to describe movement, in bacteria as well as other organisms, in response to chemoattractants or repellants, such as nutrients and poisons, respectively. *E. coli* bacteria (Figure 10) are single-celled organisms, about 2 $\mu$m long, which possess up to six flagella for movement. Chemotaxis in *E. coli* has been studied extensively. These bacteria can move in basically two modes: a "tumble" mode in which flagella turn clockwise and reorientation occurs (Figure 11, left), or a "run" mode in which flagella turn counterclockwise, forming a bundle which helps propel them forward (Figure 11, right). The motors actuating the flagella are made up of several proteins. In the terms used by Berg in [4], they constitute "a nanotechnologist's dream," consisting as they do of "engines, propellers, . . . , particle counters, rate meters, [and] gear boxes." Figure 12 shows an actual electron micrograph and a schematic diagram of a flagellar motor. The signaling pathways involved

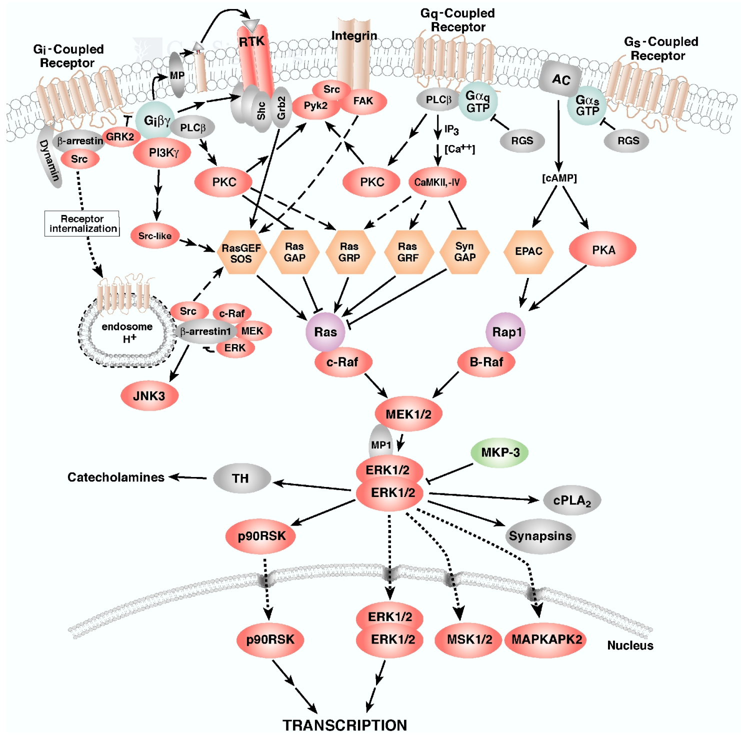


Figure 8: A GPCR pathway

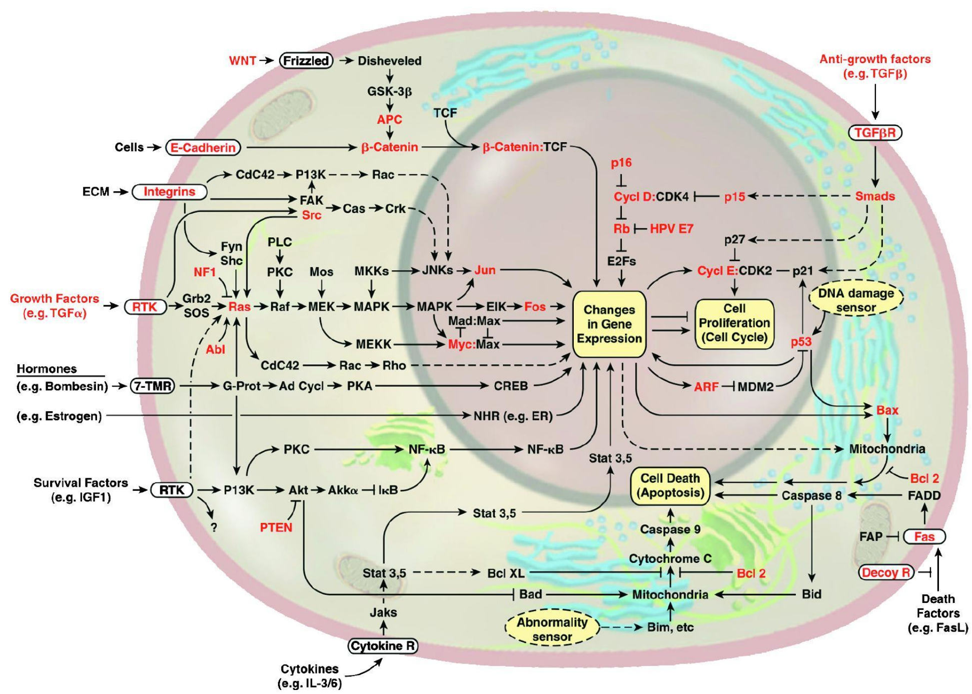


Figure 9: Signaling circuitry of the mammalian cell from [12], reprinted with permission from Elsevier

in *E. coli* chemotaxis are fairly well understood. Aspartate or other nutrients bind to receptors, reducing the rate at which a protein called CheA ("Che" for "chemotaxis") phosphorylates another protein called CheY transforming it into CheY-P. A third protein, called CheZ, continuously reverses this phosphorylation; thus,

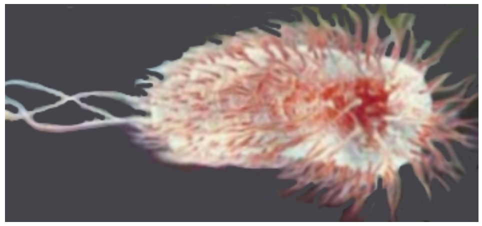

Figure 10: *E. coli* bacterium

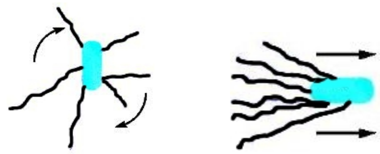

Figure 11: *E. coli* tumbling: flaggela apart. Running: flaggela in bundle

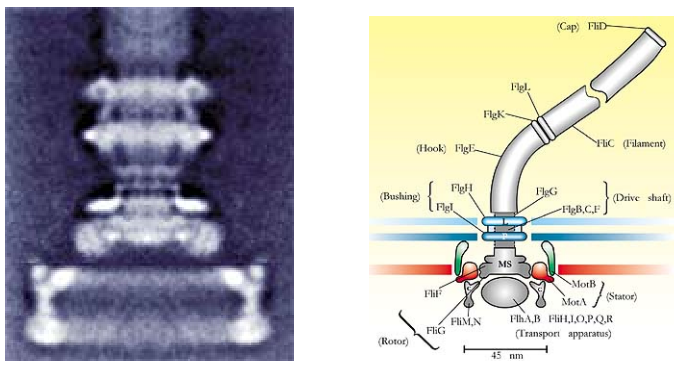


Figure 12: Electron micrograph and diagram of flagellar motor, reprinted with permission from [4]

when ligand is present, there is less CheY-P and more CheY. Normally, CheY-P binds to the base of the motor, helping clockwise movement and hence tumbling, so the lower concentration of CheY-P has the effect of less tumbling and more running (presumably, in the direction of the nutrient). A separate feedback loop, which includes two other proteins, CheR and CheB, causes adaptation to constant nutrient concentrations, resulting in a resumption of tumbling and consequent re-orientation. In this manner, *E. coli* performs a stochastic gradient search in a nutrient-potential landscape. Figure 13 shows a schematic diagram of the system responsible for chemotaxis in *E. coli*.

### 2.4 Measurement Techniques

Massive amounts of data are being generated by genomics and proteomics projects, thanks to sophisticated genetic engineering tools (gene knock-outs and insertions, PCR) and measurement technologies (fluorescent proteins, microarrays, blotting, FRET). *Polymerase chain reaction (PCR)* is a technique that amplifies DNA (typically a gene or part of a gene). Creating multiple copies of a piece of DNA, which would otherwise be present in too small a quantity to detect, PCR enables the use of measurement techniques. Let us briefly discuss a couple of these measurement technologies, in order to provide an idea of their power as well as their severe limitations.

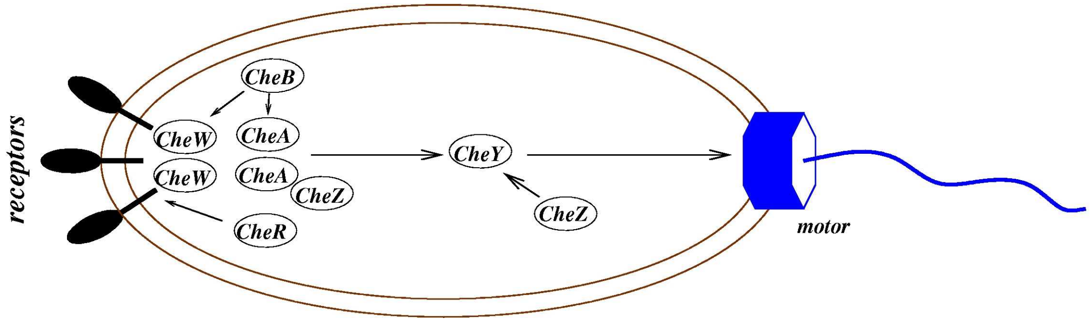


Figure 13: *E. coli* chemotactic circuit

Suppose that we wish to know at what rate a certain gene X is being transcribed under a particular set of conditions in which the cell finds itself. Fluorescent proteins may be used for that purpose. For instance, *green fluorescent protein (GFP)* is a protein with the property that it fluoresces in green when exposed to UV light. It is produced by the jellyfish *Aequoria victoria*, and its gene has been isolated so that it can be used as a *reporter gene*. The GFP gene is inserted (cloned) into the chromosome, adjacent to or very close to the location of gene X, so both are controlled by the same promoter region. Thus, gene X and GFP are transcribed simultaneously and then translated (Figure 14), and so by measuring the intensity of the GFP light emitted one can estimate

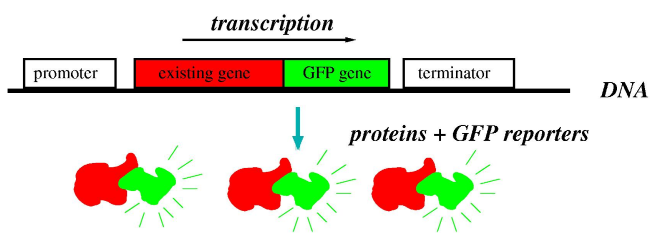


Figure 14: GFP

how much of X is being expressed.

Fluorescent protein methods are particularly useful when combined with *flow cytometry*. Flow Cytometry devices can be used to sort individual cells into different groups, on the basis of characteristics such as cell size, shape, or amount of measured fluorescence, and at rates of up to thousands of cells per second. In this manner, it is possible, for instance, to count how many cells in a population express a particular gene under a specific set of conditions.

A set of technologies collectively referred to as *gene arrays* (DNA chips, DNA microarrays, Affymatrix gene chips) provide high-throughput methods for simultaneously monitoring the activity levels of thousands of genes, thus providing a snapshot of the current gene expression activity of a cell (Figure 15). An array is built using robotics and imaging equipment, very much as in electronic chip fabrication. The array has in each location $(i, j)$ a detector "tuned" to a particular gene or small sequence of nucleotides $\mathrm{X}_{ij}$. This detector (the usual name is a "target") is the complement $\overline{\mathrm{X}}_{ij}$ of $\mathrm{X}_{ij}$ or, more likely, of a subsequence of $\mathrm{X}_{ij}$. (More precisely, one wants to find out how much of a specific X's mRNA is being transcribed. The first step is to reverse-transcribe RNA to DNA, which becomes complementary DNA (cDNA), and then PCR-amplify it. We omit details here, since we only want to explain the basic principle.) Because of hybridization, that is, the A-T and G-C base pairings for DNA, $\mathrm{X}_{ij}$ should "stick" to its complement $\overline{\mathrm{X}}_{ij}$. This allows one to estimate the presence and abundance of each $\mathrm{X}_{ij}$ in a sample. In order to be able to read the information in the different array positions, the sequences $\mathrm{X}_{ij}$ being tested for are first radioactively or fluorescently tagged, so that one can simply measure how much has accumulated at each position $i, j$. Pattern recognition, machine learning, and

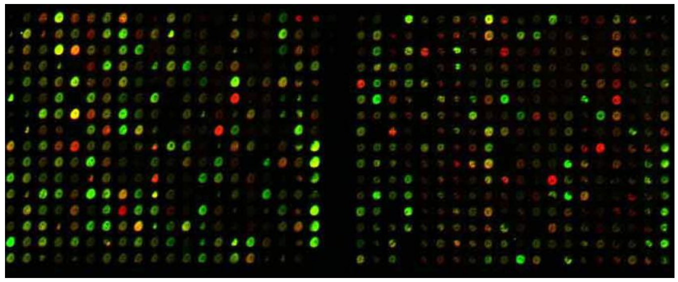

Figure 15: Gene array

control-theory tools such as clustering, Bayesian networks, and identification theory —especially when time-dependent data is available— can be and are used infer information about dynamic interactions among genes, and to sort out which particular sets of genes are triggered simultaneously or in a sequence (*co-expression* analysis) in response to different environmental factors or disease states. In control-theory language, we might think of gene arrays as giving a vector-valued output, in contrast to a technology such as GFP which provides merely a scalar value.

Actually, it is difficult to obtain absolute measurements with gene arrays, due to uncertainties in the PCR and hybridization processes. Rather, the method is often used in a comparative fashion. Gene array experiments can be done for different cell types in the same organism, for the same cell types under different experimental conditions, or even for comparing cells from two organisms, perhaps one of them having an engineered mutation of the original one. A fascinating application is the comparison of abnormal (e.g., cancerous) and normal cells, obtaining in that manner a gene expression "signature" that might be used for diagnosis.

A *Western blot* allows one to detect the presence of a specific protein, or a small number of them, in a sample taken from an experiment.[4] The proteins extracted from the sample, together with a small number of antibodies which recognize only specific proteins, are placed on membranes and allowed to interact. Different methods, for instance radioactive labeling of stains, are then used in order to visualize the results. As an example, Figure 16, taken from [20], shows Western blot data from an experiment in which three proteins (Cdc25, Wee1, and MAPK) have been observed under different conditions (concentrations 0, 25nM, etc.) of another protein named $\Delta$65-cyclin B1, during two experiments (labeled "going up" and "coming down" in the figure). The higher placements on the blot correspond in this case to the relative abundance of the phosphorylated form of the protein; for example, phosphorylated Cdc25 is more abundant in the "100" than in the "0" lanes.

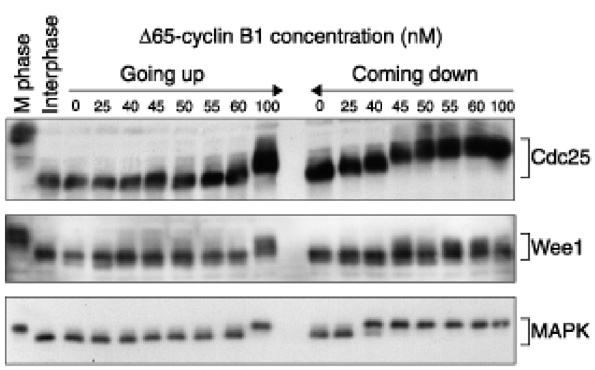


Figure 16: Western blots

[4]"Southern" blots are techniques for detecting DNA, and "Northern" blots for detecting RNA. The names originated with the first of these, which was developed by a UK biologist named Southern.

### 2.5 Limitations

Notwithstanding the power of the techniques just described, GFP, arrays, and blots, they are intrinsically noisy, because of chemical interactions in blots, production errors in arrays, or other sources of interference. In addition, the resulting measurements have low precision: very few bits of information can be extracted from data such as that shown in Figures 15 or 16. These limitations of imprecision and noise are sometimes ignored in systems biology modeling, but it is obviously pointless to try to tightly fit model parameters to such data. On the other hand, for certain types of quantities, such as the amount of calcium in a cell, currents through channels, or certain enzyme concentrations, there are other techniques that may result in higher precision measurements. In such cases, parameter fitting is more reasonable.

The field suffers from what has been called a *data-rich/data-poor* paradox: while on the one hand a huge amount of *qualitative* network (schematic modeling) knowledge is available, as evidenced by figures such as 8 and 9, on the other hand little of this knowledge is *quantitative*, at least at the level of precision demanded by most mathematical tools of analysis. The problem of exploiting this qualitative knowledge, and effectively integrating relatively sparse quantitative data, is among the most challenging issues confronting systems biology.

### 2.6 Model Organisms

Since many organisms follow the same basic principles, biologists have concentrated on a small number of *model systems*. This allows them to focus on specific systems, easing comparisons and facilitating sharing of research results. Different aspects may be easier to study in different model organisms (embryonic cycles in frog eggs, differentiation and development in flies, aging in worms), by taking advantage of fast breeding or speed of maturation.

As cataloged in the US National Institutes of Health website (http://www.nih.gov/science/models), the main mammalian models are the mouse and rat, and the main non-mammalian models are *S. cerevisiae* (budding yeast), *Neurospora* (filamentous fungus), *D. discoideum* (social amoebae), *C. elegans* (round worm), *D. melanogaster* (fruit fly), *D. rerio* (zebrafish), and *Xenopus* (frog). In addition, a popular plant model is *Arabidopsis* (a small flowering plant, member of the mustard family).

Most mathematical modeling, signal processing, and feedback control studies have been done specifically for one or another of these model systems.

## 3 Cells as Dynamical Systems

The term *genotype* refers to the genetic blueprint encoded in the DNA of a given individual, while *phenotype* refers to the actual observable physical manifestations of that information. A *single nucleotide polymorphism (SNP)*, that is, a change (mutation) in a single letter in an individual's DNA, may not have a phenotypical consequence, or it might have a catastrophic one, as is the case with cystic fibrosis in humans. Moreover, distinct species may be relatively close in genotype, yet be very far in other characteristics; for example, humans and chimpanzees are close to 99% genetically identical. Thus, differences in genotype can be tremendously amplified into phenotype. But, even accounting for environmental factors ("inputs" to the system), this amplification would seem to be somewhat inconsistent with the Central Dogma. After all, the mapping "genome $\mapsto$ proteome" is quite "continuous" in an intuitive sense and proteins determine the organism. One might then ask how large discontinuities arise.

One major contributing factor is that a cell behaves as a *nonlinear dynamical system*. As we discussed, proteins interact among themselves, both directly, through enzymatic action or through binding, as well as indirectly, through their control of gene expression. Each of these modes of interaction may involve *feedback* *loops*. Feedback is properly understood as a dynamic phenomenon, where quantities, such as concentrations of proteins, RNA, metabolites, and other cell substances are seen as functions of time.

Feedback in gene expression, to take one example, is critical to the cell's function ([7, 31]). A *transcription factor* is a protein that directs when –and possibly how many times– a gene is to be transcribed, by binding to DNA at a specific promoter or other regulatory region. Thus, a protein A may inhibit or enhance transcription of the RNA that codes for some other protein B, while B may in turn influence the production of A. Combinations of such influences are possible, as illustrated in Figure 17, in which proteins A and B must both be present in order for gene C to be active (an "and" gate in Boolean terms); the boxes labeled $P_A, P_B, P_{C_1}, P_{C_2}$ indicate

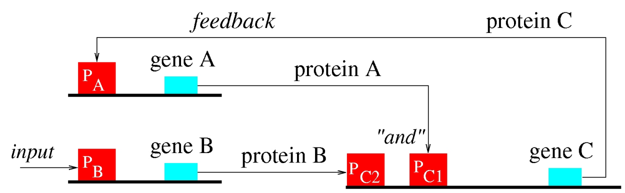


Figure 17: Proteins feed back into gene expression

regulatory sites.

The various modes of interaction are closely related: an enzymatic signal transduction network may direct the activation of a transcription factor, or a reaction of protein *dimerization* —binding of two proteins to each other— may be required for transcription factor activation. For example, the diagram in Figure 18 shows a

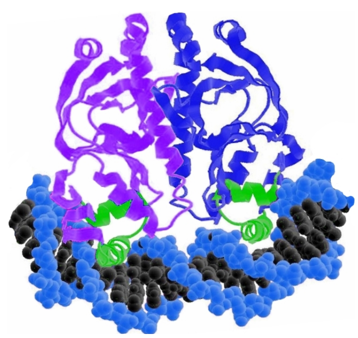

Figure 18: A homodimer, bound to DNA

*homodimer* —that is, a dimer consisting of two proteins of the same type, in this case catabolite gene activator protein (CAP), which is one of over 300 transcription factors in *E. coli*— bound to DNA, which in turn helps RNA polymerase bind and initiate transcription.

Before continuing, let us very briefly digress to mention two important sources of nonlinearities. One of them is dimerization. If a dimer consisting of a molecule of P and a molecule of Q plays a role in a reaction, then we must keep track of the amount of the dimer, let us call it D. Now, a molecule of D forms whenever a molecule of P interacts with a molecule of Q. Assuming that the medium is well-mixed —for instance due to Brownian motion— the probability that such an interaction will occur is proportional to the product of the concentrations of P and of Q. Thus, in a differential equation model that keeps track of concentrations, a product term $p(t)q(t)$ will be required in order to represent this dimerization. In particular, for homodimers one may expect to see a term like $p(t)^2$. In general, one calls an exponent appearing in this fashion a *cooperativity* index. Even higher order monomials may appear; for example, it is known that receptors in *E. coli* tend to aggregate in large numbers. Another important way in which nonlinearities appear is through saturation effects, for instance if an enzyme E catalyzes the conversion of a substrate S into a product P and the enzyme is in short supply, there will be a maximal speed at which the reaction can take place.

### Bifurcations

We wish to argue that dynamical phenomena are a main contributing factor to the appearance of discontinuities. Mathematically, such discontinuities are described as bifurcations, where a small change in a parameter results in completely different steady state behavior. (Another possibility, probably less important in this context, is the existence of chaotic dynamics, which exhibit sensitive dependence to initial conditions, so that small differences in initial states result in quickly diverging trajectories, even on finite time intervals.) Let us give a simplified illustration of the biochemical role of bifurcations; much more complicated, but totally analogous, mathematical models appear, for example, in papers dealing with embryonic development or signaling pathways. Suppose that $p(t)$ denotes the dimensionless concentration ($0 \leq p \leq 1$), at time $t$, of the protein product P of some gene, whose presence results in some observable characteristic of the individual, and that $p$ evolves in time according to the following differential equation:

$$\frac{dp}{dt} \;=\; p^2(1-p) - kp\,.$$

The negative term corresponds to degradation, and the first term to formation by an autocatalytic process, with the square term representing a dimerization. The parameter $k$ represents the activity of some enzyme that facilitates the degradation of $p$. Let's assume that $p(0) = 0.5$, an initial condition that might have been set up by another process. (In embryonic development, some of the initial conditions are set by chemical gradients placed by the mother on the fertilized egg, cf. [30].) If $k > 1/4$, then $f(p) = p(-p^2 + p - k)$ is always negative, so $p(t) \to 0$ as $t \to \infty$, that is, complete degradation of $p$ results. On the other hand, if $k < 1/4$, then $f(p) = p(-p^2 + p - k)$ has two roots $p_- < 0.5 < p_+$, so $p(t) \to p_+$. Thus *a slight perturbation of the parameter $k$ will have a drastic, discontinuous, effect on the phenotype.*

### Activation and Inhibition

It is common to classify biochemical interactions as negative (inhibitory) or positive (activating). Suppose that we consider two interacting chemicals P and Q. The rate of change of P may be affected by the concentration of Q in several different ways. For example, Q might be an enzyme that helps catalyze the production of P, or a protein whose presence triggers the expression of the gene that produces P; in this case, we say that the effect of Q on P is *positive* on P, or that Q is an *activator* of P. Alternatively, Q might be an enzyme that helps degrade P, or a protein that represses the gene that produces P, in which case we say, instead, that Q has a *negative* effect on P, or that Q *inhibits* P. Similarly, one can define activation or inhibition of Q by P. Of course, it could happen that the effect of P on Q (or of Q on P) is ambiguous, and depends on the actual concentrations of P and Q, or even of other species. However, it is often —though certainly not always— the case that biochemical models are *sign-definite*, by which we mean that this change of sign cannot happen: either P always inhibits Q or P always activates Q. As an example, take Figure 9, part of which we provide a closer look of in Figure 19. In this picture, the arrows "$\rightarrow$" indicate activation, and the symbols "$\dashv$" indicate inhibition.

To give precise definitions, one needs to settle upon a type of model for the concentrations $p(t)$ and $q(t)$ as functions of time $t$: ordinary or partial differential, probabilistic, Boolean, or hybrid equations. For concreteness, suppose that the pair of ordinary differential equations

$$\begin{aligned} \dot{p} &= f(p,q) \\ \dot{q} &= g(p,q) \end{aligned}$$

adequately describes the interaction between P and Q (as usual in control theory, using dot for time derivatives and omitting "$t$" arguments). Then, Q is an activator of P if the partial derivative $\frac{\partial f}{\partial q}(p,q)$ is positive, or at least nonnegative, everywhere, and Q is an inhibitor of P if this derivative is negative, or at least nonpositive,

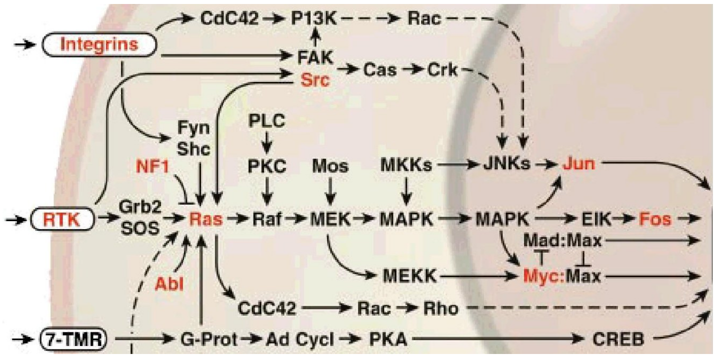


Figure 19: Zooming-in on Figure 9

everywhere. The non-sign-definite case would be that in which the partial derivative is positive for some values of the state variables $(p, q)$ and negative for others, as with the equation $\dot{p} = (1-p)q$, where $\frac{\partial f}{\partial q} = 1 - p$ is positive if $p < 1$ and negative if $p > 1$.

### Two-Species Interactions

As the number of proteins and other species increase, the complexity of feedback loops and dynamics exhibited by biochemical networks can be, in principle, quite arbitrary. However, some of the main behaviors in which biologists have focused their interest arise already in systems that involve just two interacting chemicals, Figure 20.

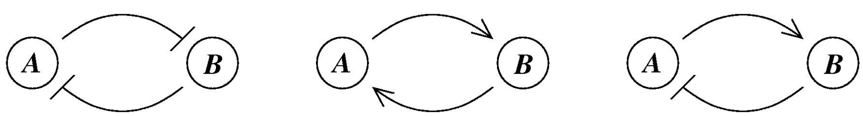


Figure 20: Mutual inhibition. Mutual activation. Activation-inhibition

In *mutual inhibition*, each species inhibits the other one. If some external input signal helps to transiently increase the concentration of A sufficiently over that of B, then A will repress B. Since B is at a low concentration, it will not repress A. Assuming that A can maintain its high level —due for example to some autocatalytic reaction, or to the influence of other variables not shown— this situation will persist until such a time when some other external factor allows B to gain an upper hand over A. The system will, therefore, memorize which of the two components, A or B, was last activated externally; this "toggle-switch," analogous to similar ones found in electronic devices, plays a central role in differentiation and other biological forms of memory. See for instance older work on the lambda phage lysis-lysogeny switch and the hysteretic *lac* repressor system [19, 21], as well as more recent references such as [10, 5, 23, 3].

In *mutual activation*, each species activates the other. Now, if some external input signal helps to transiently increase the concentration of A, then B will be activated by A, and B will, in turn, enhance A even more. In effect, a sufficiently large external signal, applied to either A or B, results in a large increase in both A and B. (If the signal is not strong enough, we will assume that A and B stay small.) This mode of positive feedback appears in biomolecular systems that amplify signals, as well as systems that produce a "binary" response to external stimuli, and it is thought to play a role in cell decision-making.

Finally, a net negative feedback as in *activation/inhibition* loops is, as usual in the field of control theory, the mechanism responsible for set-point regulation, or as biologists say, *homeostasis*. It plays a role also in turning signals "off" after activation: many cell signals are too expensive metabolically to be maintained at a high level.

These behaviors are associated with different phase-space pictures, which we discuss now, for concreteness, for ordinary differential equation models.

### 3.1 Phase Spaces and Step Response

Three of the main types of phase-space behaviors that have attracted particular attention from biologists studying biomolecular dynamics are: systems with a unique stable state, systems with multiple attracting states, and limit cycle oscillators, cf. Figure 21. These three types of behaviors are intimately linked, and often give rise to

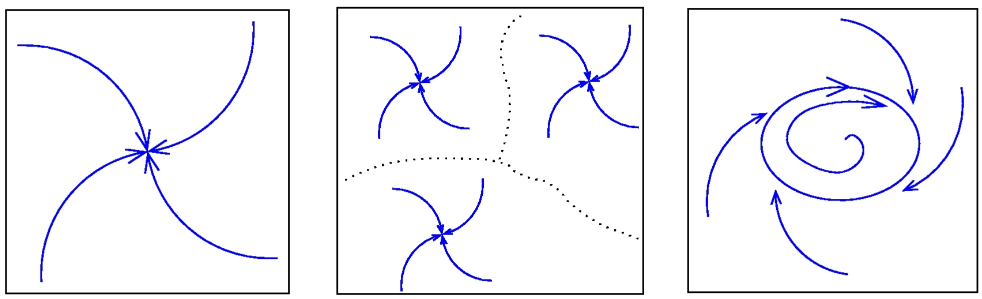

Figure 21: One or multiple steady states; Limit cycles

each other, as we will discuss.

Uniqueness of steady states, and globally asymptotic stability, are quite common among simple biochemical reactions, although it is not always easy to prove theorems insuring this behavior (we discuss some such results later). Systems with multiple attractors arise in many forms, a typical one of which is the interaction between two processes, such as formation and degradation, each of which by itself would lead to global stability. Relaxation, or hysteresis-driven, oscillators are those in which to a system with multiple attractors one adds a slow parameter adaptation law. Other oscillators arise through a Hopf bifurcation phenomenon –basically an unstable linear oscillator, plus a nonlinear term that prevents escape to infinity and thus confines trajectories– from negative feedback loops around otherwise mono-stable systems.

The transitions (bifurcations) between qualitative behaviors such as mono- and multiple-stability, or the onset of oscillations, are phenomena which frequently arise when parameters in systems are modified. In molecular biology modeling, a parameter may typically represent a concentration of an external ligand, a voltage applied to a voltage-gated channel, the concentration of a signaling molecule (as an input to a cellular subsystem), an enzyme concentration affecting a reaction, or the degree of effective cooperativity (Hill coefficient) of a reaction.

For example, suppose that the rate of change of the concentration of some substance $p$ has the following form:

$$\dot{p} \;=\; \frac{V_{max}u}{k_m + u} \;-\; kp$$

where we fix the parameters $V_{max}$, $k_m$, and $k$, and where $u$ is a parameter, not fixed yet, which might correspond, for instance, to the concentration of substrate that is used in making $p$. The term $-kp$ is a degradation term, while the first term is a *Michaelis-Menten* formation term. (Michaelis-Menten kinetics are really a singular perturbation reduction of a more complicated underlying enzymatic reaction, see e.g. [9] for details.) Note that $V_{max}$ is the maximum possible speed of the formation reaction, while $k_m$ ("$m$" for middle) is the concentration of $u$ for which the rate happens to be $V_{max}/2$. Now assume that $u$ is fixed at some value $u_0$. The concentration $p(t)$ will then, from any initial condition $p(0)$, converge to the steady state $p_0 = \frac{(V_{max}/k)u_0}{k_m+u_0}$. In a typical set of experiments, a biologist or biochemist will set the concentration to a given value $u_0$, let the system relax to the corresponding steady state $p_0$, and repeat for various values of $u_0$, thus obtaining a plot of $p_0$ against different such $u_0$'s. In Figure 22 we show the plot (with $V_{max} = 1$, $k_m = 0.25$, $k = 1$) for the above example.

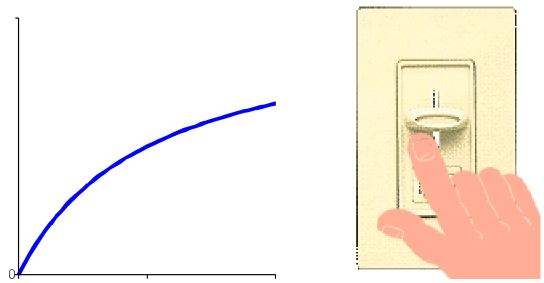


Figure 22: Hyperbolic steady-state response

We'll call this graph, using terminology borrowed from control-theory, the *steady state response to step inputs*, where we think of $u_0$ as the magnitude of a constant input applied to the system. Depending on the context, this plot might be called a *dose-response curve* or *receptor activity plot* when $u$ represents a concentration of ligand and $p$ the level of some indicator of receptor activity, a *steady-state phosphorylation level* plot when $u$ represents a signal that affects the phosphorylation level of a protein, and so forth. The response in this example is *graded* in the sense that it is proportional to the parameter $u_0$, at least over a large range of values $u_0$, even though it eventually saturates. It is said to be a *hyperbolic* response, in contrast to a *sigmoidal* response as in Figure 23. A sigmoidal response arises typically from a reaction such as:

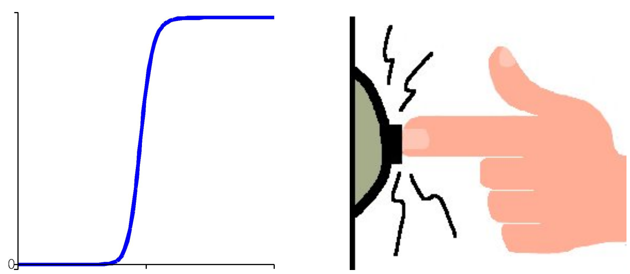


Figure 23: Sigmoidal steady-state response

$$\dot{p} \;=\; \frac{V_{max}u^r}{k_m^r + u^r} \;-\; kp$$

where the *Hill coefficient* $r$ is greater than one (in our figure, we used $r = 20$, $V_{max} = 1$, $k_m^r = 0.4$, and $k = 1$). The parameter $r$ is a cooperativity index. The sharp increase, and saturation, means that a value of $u_0$ which is under some threshold (roughly, $u < k_m$) will not result in an appreciable result ($p_0 \approx 0$, in steady state) while a value that is over this threshold will give an abrupt change in result ($p_0 \approx V_{max}/k$, in steady state). While the first example, when we think of $u_0$ as displacement of a slider or button, is analogous to the behavior of a light-dimmer, the second one is closer to that of a doorbell. (We do not define here precisely the difference between sigmoidal and hyperbolic responses. One possible definition is in terms of inflection points in the graph. But there is no need to be formal, since we want to keep the discussion intuitive at this point.)

Sigmoidal responses are characteristic of many signaling cascades, which display what biologists call an *ultrasensitive* response to inputs. If the purpose of a signaling pathway is to decide whether a gene should be transcribed or not, depending on some external signal sensed by a cell, for instance the concentration of a ligand as compared to some default value, such a binary response is required. Cascades of enzymatic reactions can be made to display ultrasensitive response, as long as at each step there is a Hill coefficient $r > 1$, since the derivative of a composition of functions $f_1 \circ f_2 \circ \ldots \circ f_k$ is, by the chain rule, a product of derivatives of the functions making up the composition ([11]). Thus, the slopes get multiplied, and a steeper nonlinearity is produced. In this manner, a high effective cooperativity index may in reality represent the result of composing several reactions, perhaps taking place at a faster time scale, each of which has only a mildly nonlinear behavior.

In practice, steady-state step response curves are interpolated from a number of measurements taken for various values of $u_0$. For a concrete, although relatively old, example, we show in Figure 24, taken from [8], a (log scale) plot of the degree of cAMP receptor modification after 15 minutes of constant exposure to the

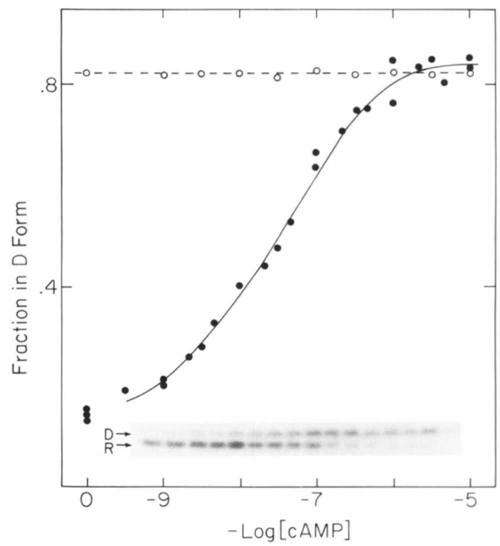


Figure 24: Example of steady-state response, from [8]

stimulant cAMP, in *Dictyostelium*. The locations of the black circles are obtained by reading the Western blots shown in the inset.

We mentioned that systems with multiple attractors sometimes arise through the interaction of formation and degradation processes. A typical way in which this happens is as follows. Suppose that the output $y$ of a system, for example $y = p$ in the example that we have been considering, is fed-back into the input $u$, as shown diagrammatically in Figure 25(a). Physically, we are dealing with an autocatalytic process, and may


Figure 25: (a) Feed-back $u = y$; (b) Feed-back $u = g \cdot y$

think simply of $u$ being equal to $p$ —this could happen for example if $p$ helps promote its own transcription— or perhaps there could be a more complicated positive feedback pathway from $p$ to $u$. Mathematically, we substitute $u = y$ into $\dot{p} = \frac{V_{max} u^r}{k_m^r + u^r} - kp$ (where $r = 1$ or $r > 1$), and obtain the closed-loop equation:

$$\dot{p} \;=\; \frac{V_{max} p^r}{k_m^r + p^r} \;-\; kp\,.$$

We plot in Figure 26 both the first term (formation rate) and the second one (degradation), in cases where $r = 1$ (left) or $r > 1$ (right). Let us analyze the solutions of the differential equation. In the first case, $r = 1$, for

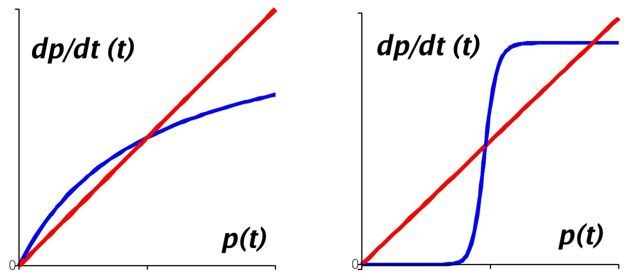


Figure 26: Bistability arises from sigmoidal formation rates

small $p$ the formation rate is larger than the degradation rate, while for large $p$ the degradation rate exceeds the formation rate; thus, the concentration $p(t)$ *converges to a unique intermediate value.* In the second case, however, the situation is more interesting: for small $p$ the degradation rate is larger than the formation rate, so the concentration $p(t)$ *converges to a low value*, but, in contrast, for large $p$ the formation rate is larger than the degradation rate, and so the concentration $p(t)$ *converges to a high value* instead. In summary, two stable states are created, one low and one high, by this interaction of formation and degradation, if one of the two terms is sigmoidal. (There is also an intermediate, unstable state.) *These facts are totally elementary, but they serve to motivate a theory based upon monotone systems, cf. [26, 25], which provides a far-reaching generalization.*

Whether, under feedback, a mono-stable or a multi-stable system results, depends on the shape of the curves, which in turn is determined by the numerical values of the parameters. For example, the hyperbolic case, shown in the left panel of Figure 26, corresponds to $r = 1$, while $r \gg 1$ tends to produce pictures like the one shown in the right panel.

Other parameters also play a role. Let us consider a situation where the strength of the feedback can be modulated in some fashion, for example due to some additional transcriptional or enzymatic control. The simplest case (the theory works equally well in more complex scenarios) is when $u$ is proportional to the output $y$: $u = g \cdot y$, where "$g$" (a "feedback gain" in engineering) is a parameter than quantifies the proportion (amplification, if $g > 1$) of $y$ that is fed back as input, That is, instead of closing the loop simply with $u = y$ as in Figure 25(a), we now wish to study the effect of a more general feedback $u = g \cdot y$, where $g \neq 1$, as in Figure 25(b).

For example, consider the sigmoidal curve shown in the left panel of Figure 27, showing the steady-state

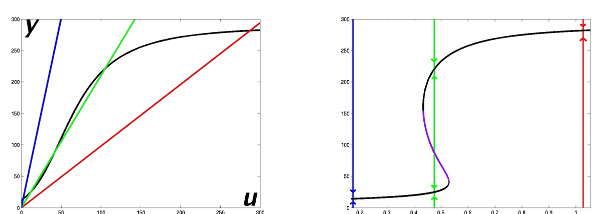


Figure 27: Open-loop step-response and three feedback gains; Corresponding bifurcation diagram

step response $y = k(u)$ of a certain system that we will study later. For any fixed value of $g$, the steady states of the closed-loop in Figure 25(b) are in one-to-one correspondence with those pairs $(u, y)$ for which both $y = k(u)$ and $u = gy$, that is to say, the intersections of the graph of $y = k(u)$ with the line $y = (1/g)u$. In particular, we show in the left panel of Figure 27 the lines $y = (1/g)u$ with slopes corresponding to the three special values $g = 1/0.98$, $g = 1/2.1$, and $g = 1/6$. The middle line would correspond to a bistable case as in the right panel of Figure 26, while the other two lines correspond to cases where a single steady state will occur, either a "low $y$" or a "high $y$" one. We may plot the $y$-coordinates of these intersections against the values of the gains $g$. Observe that this plot, the bifurcation diagram, can be easily obtained by a projective transformation from the data given by the steady state response $y = k(u)$: it is simply given, in parametric form, as the set of pairs $\left(\frac{u}{k(u)}, k(u)\right)$ parametrized by possible input values $u$. See the right panel of Figure 27. Thus, if $k$ is obtained from experimental data, the bifurcation diagram can be immediately derived from it.

An intuitive way of thinking of the dependence of the steady state on the parameter $g$ is by viewing $g$ as the force being applied on a light switch (let us say, positive means up, and negative means down) as in Figure 28. A strong enough positive force will turn the light on, and a strong enough negative force will turn it off, no

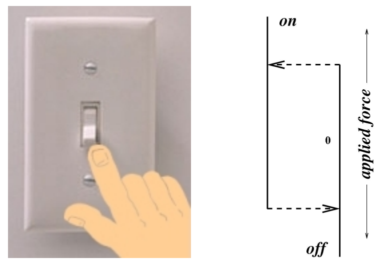


Figure 28: Hysteretic behavior

matter how we started. An intermediate value will have an effect that depends on the initial state: if the switch is only partially up, but the light is off, a small force will leave it off; if it is on, it will stay on. This is the bistable case, where the steady state attained depends on the initial state. *Hysteresis* is the term used to describe the phenomenon in which the actual steady state depends on the history of the system. One of the main roles of such hysteretic behavior is in producing oscillations. Imagine an indecisive individual, who, when the light is off starts applying a higher and higher upward pressure on the switch, but, when the light turns on, changes his mind and starts applying a downward pressure, repeating the process forever. The resulting oscillation is called a hysteresis-based or relaxation oscillation.

Biologically, relaxation oscillators appear to underlie many important cell processes. As a concrete example, let us briefly discuss the early embryonic cell cycle in frog eggs (*Xenopus* oocytes), in which there occur a set of 12 synchronous cell divisions, starting from just one cell in the fertilized egg and resulting in 4096 cells. The normal cell cycle in a mature organism involves several steps: mitosis (M, the actual cell division) and interphase, the latter made up of the substeps Gap1 (G1, when the cell grows, in preparation for cell division), synthesis (S, when DNA is replicated), and Gap2 (G2, a second gap before returning to M). In the early embryo, though, there are no checkpoints (stopping at gaps), and the cell divisions take place in quick succession. The divisions are controlled by proteins named *cyclin-dependent kinases (Cdk's)*, so called because they are active when cyclins (another type of protein) are bound to them. Examples of Cdk's are *cell-division cycle (Cdc)* proteins. A dimer made up of one of these, Cdc2, and cyclin B, a type of cyclin, is called *mitosis promoting factor (MPF)*. MPF can be in four different phosphorylation states, depending on the binding at the amino acid in position 167 (which is a threonine, and hence is referred to as "threonine-167") by the protein kinase CAK, and at a tyrosine-15 site by another protein called Wee1 when the latter is non-phosphorylated. The phosphorylation of MPF at the tyrosine-15 site is reversed by yet another protein called Cdc25, which acts in that manner when it is phosphorylated. The active form of MPF is that in which *only* the threonine-167 has been phosphorylated. When MPF is active, it phosphorylates both Wee1 and Cdc25. Leaving aside, for simplicity, the action of CAK and two of the phosphorylation states, the reactions between MPF, Wee1, and Cdc25 are as shown in Figure 29. (See [14] for more details.) This system is a net positive feedback system. One can give an ordinary differential equation model, and appropriate parameter values, so that there are two possible steady states: one corresponding to mitosis (high concentration of activated MPF as well as Wee1-P

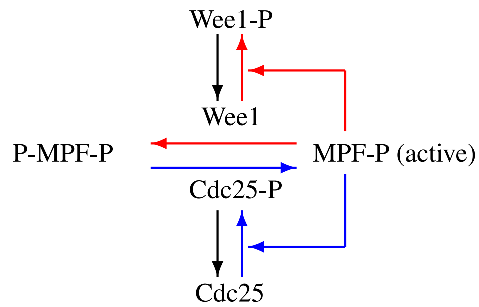


Figure 29: Cell cycle subsystem

and Cdc25-P) and one to interphase (higher concentration of inactivated MPF, Wee1, and Cdc25). Depending on the total amount of MPF (adding active and inactive forms), these two states may theoretically exist in the same system (bistable regime) or only one of them may be possible.

It is believed that the oscillations are produced by a relaxation oscillation mechanism: the concentration of cyclin B can be viewed as a parameter which controls the concentration of MPF. Through a negative feedback loop involving yet other players (not shown), cyclin B is degraded when MFP is activated, making the system move between the monostable and bistable regimes, much as with the light-switch example. How does one test this hypothesis? In a beautiful experimental demonstration, Joe Pomerening and Jim Ferrell at Stanford blocked the degradation of cyclin B by introducing instead a mutated form which cannot be degraded. In effect, this broke the negative feedback loop and left the system in Figure 29 isolated. To verify that this system is indeed bistable, they manipulated the concentration of cyclin B and let the system relax to steady state. If the system is indeed bistable, a bifurcation diagram like the one shown in the right panel of Figure 27 should result. Indeed, the results, shown in the Western blots in Figure 16, indicate just this hysteretic behavior ("going up" versus "coming down" in parameter space). Further confirmation of this bistable behavior was obtained from morphological observations. Figure 30 shows, for different values of the parameter and at steady-state, observations done under a microscope. (We omit details of how this was done, which is an interesting story in itself.) The pictures for parameter values between 40 and 60 show the two possible steady states in this bistable system, each of which is arrived at depending on the history of the system. As the parameter is slowly increased from 0 to 100, starting in interphase, we see nuclei that stay well-formed, indicating interphase, for a large range of parameters, while M phase (nuclear envelope broken down, chromosomes condensed) is only observed for the value 100. Conversely, going down, the M phase view persists for a large range of parameters.

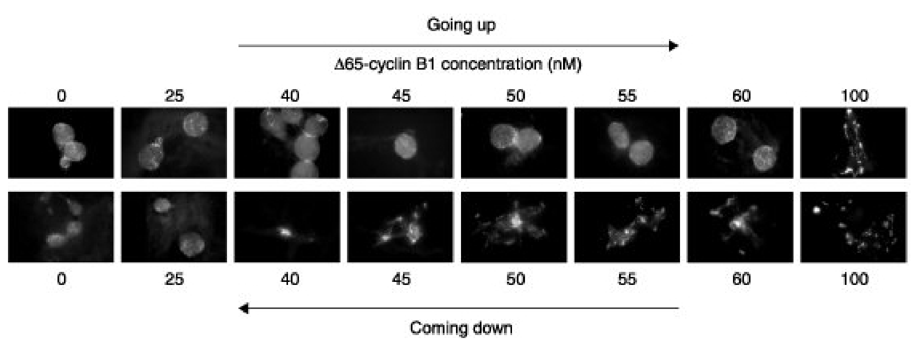


Figure 30: Hysteresis and bistability seen under a microscope, from [20]

# 4 An Example

We close this quick introduction with an example of a model which has appeared in several recent research papers.

*Mitogen-Activated Protein Kinase (MAPK) cascades* are a ubiquitous "signaling module" in eukaryotes, involved in proliferation, differentiation, development, movement, apoptosis, and other processes ([13, 16, 29]) Figures 8 and 9 show MAPK cascades as subsystems. There are several such cascades, as diagrammed in Figure 31, and they share the property of being composed of a cascade of three kinases, see Figure 32 (see [6]

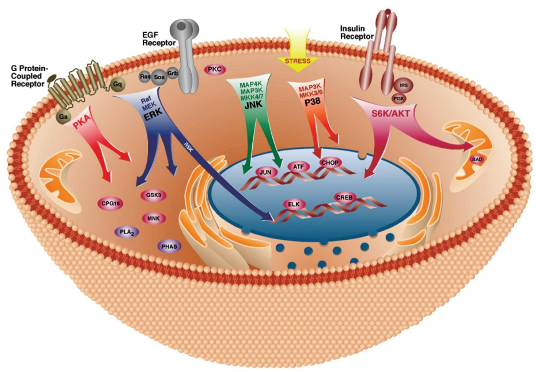


Figure 31: MAPK Cascades, reprinted by permission from [22]

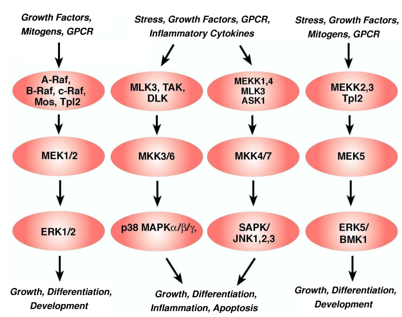


Figure 32: Different MAPK cascades, similar structure.

for several similar illustrations). The basic rule is that two proteins, called generically MAPK and MAPKK (the last K is for "kinase of MAPK," which is itself a kinase), are active when doubly phosphorylated, and MAPKK phosphorylates MAPK when active. Similarly, a kinase of MAPKK, MAPKKK, is active when phosphorylated. A phosphatase, which acts constitutively (that is, by default it is always active) reverses the phosphorylation. There are many models of MAPK cascades, with varying levels of complexity. We pick here the Huang-Ferrell model ([13, 2]), see Figure 33, and use $z_1(t)$ for MAPK concentration, $z_2(t)$ for the concentration of the singly-phosphorylated MAPK-P, and so forth. The simplest assumptions about the dynamics are made. For example, take the reaction shown in the square in Figure 33. As $y_3$ (MAPKK-PP) facilitates the conversion of $z_1$ into $z_2$ (MAPK to MAPK-P), the rate of change $\dot{z}_1$ should include a term $-\alpha(z_1, y_3)$ (and $\dot{z}_1$ has a term $-\alpha(z_1, y_3)$) for some (otherwise unknown) function $\alpha$ such that $\alpha(0, y_3) = 0$ and $\frac{\partial \alpha}{\partial z_1} > 0$, $\frac{\partial \alpha}{\partial y_3} > 0$ (more enzyme or more substrate results in a faster reaction, but nothing happens if there is no substrate). There will also be a term $+\beta(z_2)$ to reflect the phosphatase action.

Because the the variables $y_i$ and $z_i$ represent, respectively, different forms of the same protein, there are "conservation laws" (much like conservation of mass in chemistry or of energy in physics) that help simplify

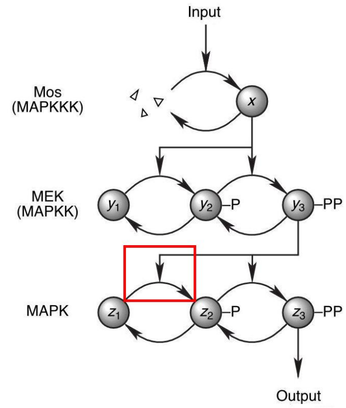


Figure 33: A MAPK system

the equations. Let us write $y_1(t) + y_2(t) + y_3(t) \equiv y_{\text{tot}}$ (total MAPKK) and $z_1(t) + z_2(t) + z_3(t) \equiv z_{\text{tot}}$ (total MAPK), and eliminate the variables $y_2$ and $z_2$. Once we do this, and write $y_2 = y_{\text{tot}} - y_1 - y_3$ and $z_2 = z_{\text{tot}} - z_1 - z_3$, we are left with the variables $x, y_1, y_3, z_1, z_3$. For instance, the equations for $z_1, z_3$ look like:

$$\begin{aligned}\dot{z}_1 &= -\alpha(z_1, y_3) + \beta(z_{\text{tot}} - z_1 - z_3) \\ \dot{z}_3 &= \gamma(z_{\text{tot}} - z_1 - z_3, y_3) - \delta(z_3)\end{aligned}$$

for appropriate functions $\alpha, \beta, \gamma, \delta$. The equations for the remaining variables are similar.

Positive and negative feedback loops around MAPK cascades have been a topic of interest in the biological literature. For example, in [10] one finds the study of positive feedback (on Mos by ERK) in the context of progesterone-induced oocyte maturation in frogs. For another example, [15] and [24] looked at negative feedback, also in an ERK cascade, but affecting upstream proteins. See [26, 25] for discussions of dynamics of MAPK cascades under positive and negative feedback, and more references to the literature.

# 5 New Problems in Systems and Control Theory

In [26] —see also the previous paper [25]— we argue that questions in systems biology often differ in an essential manner from similar-sounding questions in engineering applications, thus leading one to entirely new theoretical control and systems theory problems. We also describe in that paper a topic in which the author has recently worked, namely, the combination of network-like, qualitative knowledge, with a comparatively small amount of quantitative data, in order to help characterize global behavior.

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